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Some theory of ordinary generating functions (Ardila §2.2)

Roughly speaking, if A is a class of combinatorial structures, w/ $a_n = \#$ (weighted?) A -structures of "size" n ($\in$ ring R), then we can form the ordinary generating function $A(x) = \sum_{n \geq 0} a_n x^n \in R[[x]]$

Prop. • If C structures of size n are a choice of either an A -structure or B -structure of size n (" $C = A + B$ ") (i.e., $c_n = a_n + b_n$) then $C(x) = A(x) + B(x)$.

• If C structures of size n are a choice of
 - an A -structure of size i
 - a B -structure of size j (" $C = A \times B$ ")
 for some $i+j=n$ (i.e., $c_n = \sum_{i+j=n} a_i b_j$)
 then $C(x) = A(x) \cdot B(x)$.

• If C -structures of size n are a choice of B -structures of sizes $i_1, i_2, \dots, i_k$ for some $i_1 + i_2 + \dots + i_k = n$, $i_j \geq 0$, for some $k \geq 0$

(i.e., $c_n = \sum_{(i_1, i_2, \dots, i_k)} b_{i_1} b_{i_2} \dots b_{i_k}$) (" $C = \text{Seq}(B)$ ")
 $\downarrow$
 "sequences of B -structures"

then $C(x) = \frac{1}{1-B(x)}$.

Pf of the proposition is straightforward (just uses definition of addition + multiplication of FPS's).

Now let's see some examples of how to apply this...

EXAMPLES (see also Ardila § 2.2.2)

① (Partitions w/ bounded part sizes)

Let $P_{\leq k}(n) := \#\{\text{partitions } \lambda = (\lambda_1, \lambda_2, \dots) \vdash n : \lambda_i \leq k\}$,

i.e., partitions of n into parts of size at most k .

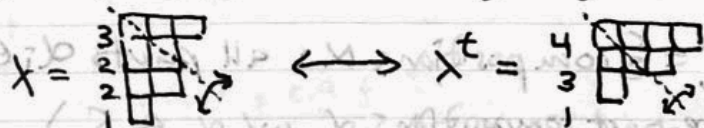
$$\text{Then } P_{\leq k}(q) = \sum_{n \geq 0} P_{\leq k}(n) q^n = \sum_{\lambda: \lambda_i \leq k} q^{|\lambda|}$$

$$= \underbrace{\frac{1}{1-q}}_{\substack{\text{o.g.f. for } \lambda \\ \text{w/ only parts} \\ \text{of size } = 1}} \cdot \underbrace{\frac{1}{1-q^2}}_{\substack{\text{o.g.f. for } \lambda \\ \text{w/ only parts} \\ \text{of size } = 2}} \cdot \dots \cdot \underbrace{\frac{1}{1-q^k}}_{\substack{\text{o.g.f. for } \lambda \\ \text{w/ only} \\ \text{parts of size } = k}} = \frac{1}{(1-q)(1-q^2)\dots(1-q^k)}$$

↑
we saw the
 $k \rightarrow \infty$ limit
of this before

i.e., $\mathcal{C} = \{\lambda : \lambda_i \leq k\} = \text{Seq}(\text{ones}) \times \text{Seq}(\text{Twos}) \times \dots \times \text{Seq}(k\text{'s})$

Remark The conjugate (or transpose) of a partition λ , denoted λ^t , is the partition whose Young diagram is the reflection of the Young diagram of λ across 'main diagonal'.



Conjugation is a bijection (in fact, involution!) on partitions, and $\lambda_1 = \ell(\lambda^t)$ ← recall length $\ell = \#$ of parts.

Hence also $P_{\leq k}(n) = \#\{\text{partitions } \lambda \vdash n : \ell(\lambda) \leq k\}$.

And so we also have $\sum_{\lambda: \ell(\lambda) \leq k} q^{|\lambda|} = \frac{1}{(1-q)(1-q^2)\dots(1-q^k)}$

Similarly, $\sum_{\lambda: \lambda_1 \leq k} q^{|\lambda|} t^{\ell(\lambda)} = \frac{1}{(1-tq)(1-tq^2)\dots(1-tq^k)} = \sum_{\lambda: \ell(\lambda) \leq k} q^{|\lambda|} t^{\lambda_1}$

② (Compositions)

Recall $c(n) = \# \text{ compositions } \alpha \models n$.

$$\text{We have } \sum_{n \geq 0} c(n) x^n = \frac{1}{1 - (x + x^2 + x^3 + \dots)} = \dots = 1 + \frac{x}{1 - 2x}.$$

i.e., " $\mathcal{C} = \text{Seq}(\text{one-part compositions})$ "

o.g.f. for compositions of n w/ one part, since $\exists$ unique such comp.

↑ saw this previously

③ (Partitions/compositions w/ restricted parts)

Generalizing previous two examples, let $S \subseteq \{1, 2, 3, \dots\}$ be any subset of positive integers.

First consider $\mathcal{P} = \{ \text{partitions } \lambda : \text{all parts } \lambda_i \in S \}$.

Then $\mathcal{P} = \prod_{j \in S} \text{Seq}(j\text{'s})$, so

$$C(x) = \sum_{n \geq 0} \# \{ \lambda \vdash n : \text{all } \lambda_i \in S \} x^n = \prod_{j \in S} \frac{1}{1 - x^j}.$$

Next consider $\mathcal{C} = \{ \text{compositions } \alpha : \text{all parts } \alpha_i \in S \}$.

Then $\mathcal{C} = \text{Seq}(\text{one-part compositions } \alpha_i \text{ w/ } \alpha_i \in S)$, so

$$C(x) = \sum_{n \geq 0} \# \{ \alpha \models n : \text{all } \alpha_i \in S \} x^n = \frac{1}{1 - \sum_{j \in S} x^j}.$$

e.g., if $S = \{1, 2\}$, then

$$\sum_{n \geq 0} \# \{ \text{comp. of } n \text{ into } 1\text{'s and } 2\text{'s} \} x^n = \frac{1}{1 - (x + x^2)},$$

which we saw on 1st week of class, with the Fibonacci numbers!

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④ (Stirling #'s of the 2nd kind)

DEFN A set partition of $[n]$ is a set $\Pi = \{\pi_1, \pi_2, \dots, \pi_k\}$ of subsets $\pi_i \subseteq [n]$ s.t.

- (nonempty) $\pi_i \neq \emptyset \forall i$
- (disjoint) $\pi_i \cap \pi_j = \emptyset \forall i \neq j$
- (covering) $\bigcup \pi_i = [n]$

The π_i are called the blocks of the set partition Π .

DEFN $S(n, k) := \#$ of set partitions of $[n]$ into exactly k blocks
 "Stirling #'s of the 2nd kind"

e.g. $1|2|3 \quad | \quad 12|3, 13|2, 23|1 \quad | \quad 1 \ 2 \ 3$
 $S(3,3)=1 \quad S(3,2)=3 \quad S(3,1)=1$

Table of $S(n, k)$:

$n \backslash k$	1	2	3	4
1	1	0	0	0
2	1	1	0	0
3	1	3	1	0
4	1	7	6	1

(Pascal-like) Recurrence for $S(n, k)$:

$$S(n, k) = S(n-1, k-1) + k \cdot S(n-1, k) \quad \text{for } k > 1$$

$\underbrace{S(n-1, k-1)}_{\substack{n \text{ is in a single} \\ \text{block}}} + \underbrace{k \cdot S(n-1, k)}_{\substack{n \text{ goes into} \\ \text{one of the} \\ k \text{ other blocks}}}$

w/ initial conditions $S(n, 1) = 1 \forall n$
 $S(0, 0) = 1, S(n, k) = 0$ if $k > n$

Let's study the o.g.f. $F_k(x) = \sum_{n \geq 0} S(n, k) x^n$ in 2 ways:

① Solve recurrence: for $k \geq 2$,

$$\sum_{n \geq 0} S(n, k) x^n = \sum_{n \geq 0} S(n-1, k-1) x^n + \sum_{n \geq 0} k \cdot S(n-1, k) x^n$$

$$F_k(x) = x F_{k-1}(x) + k x F_k(x)$$

$$(1 - kx) F_k(x) = x F_{k-1}(x)$$

$$F_k(x) = \frac{x}{1 - kx} F_{k-1}(x)$$

(and for $k=1$, $F_1 = \sum_{n \geq 0} S(n, 1) x^n = x + x^2 + x^3 + \dots = \frac{x}{1-x}$)

$$\Rightarrow F_k(x) = \frac{x}{1-kx} \cdot \frac{x}{1-(k-1)x} \cdots \frac{x}{1-2x} \cdot \frac{x}{1-x} = \frac{x^k}{(1-x)(1-2x)\cdots(1-kx)}$$

(b) Let $A_m :=$ the structure of strings of letters from $[m]$ that start w/ an m , whose size is length

(e.g. $m=3$ $\frac{31312}{\text{size } 5}$ or $\frac{3311}{\text{size } 4}$)

Prop. $\left\{ \begin{array}{l} \text{Set partitions} \\ \pi \text{ of } [n] \text{ w/ } k \text{ blocks} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{(total) size} = n \text{ structures} \\ \text{in } A_1 \times A_2 \times \dots \times A_k \end{array} \right\}$

$\pi \mapsto$ the restricted growth function $f: [n] \rightarrow [k]$ associated to π

E.g. $n=16$ $k=4$

	①	②	③	④	i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
π	1	2	4	5	8	12	3	6	9	10	7	11	16	13	15						
$f(i)$	0	0	2	0	0	2	3	0	2	3	0	2	3	1	4	2	3	1	4	2	3
	$\in A_1$	$\in A_1$	$\in A_3$	$\in A_1$	$\in A_1$	$\in A_3$	$\in A_4$	$\in A_1$	$\in A_3$	$\in A_4$	$\in A_1$	$\in A_3$	$\in A_4$	$\in A_2$	$\in A_4$	$\in A_1$	$\in A_3$	$\in A_2$	$\in A_4$	$\in A_2$	$\in A_3$

number the blocks of π ①, ②, ..., ④ according to increasing smallest elements

$f(i) :=$ block # containing i

Pf: Exercise for you

Cor $F_k(x) = \frac{x^1}{1-x} \cdot \frac{x^2}{1-2x} \cdot \dots \cdot \frac{x^k}{1-kx} = \frac{x^k}{(1-x)(1-2x)\dots(1-kx)}$

$x + x^2 + x^3 + \dots$ $x + 2x^2 + 4x^3 + \dots$ $x + kx^2 + k^2x^3 + \dots$

10/6 The two kinds of Stirling #'s

How are $S(n, k)$ and $C(n, k)$ related? $\# \{ \pi \in S_n : \pi \text{ has } k \text{ cycles} \}$

$S(n, k)$ is Stirling #'s of 2nd kind $C(n, k)$ is (signed) Stirling #'s of 1st kind

Note: The $C(n, k)$ satisfy a similar (Pascal-like) recurrence:

$$C(n, k) = C(n-1, k-1) + (n-1)C(n-1, k)$$

n is in a 1-cycle n maps to some $i \in [n-1]$

$n \backslash k$	1	2	3	4
1	1	0	0	0
2	1	1	0	0
3	2	3	1	0
4	6	11	6	1

But the real connection between Stirling #'s is...

Prop. (i) $x^n = \sum_{k=1}^n S(n,k) (x)_k$ where $(x)_k := x(x-1)(x-2)\dots(x-(k-1))$
 while (ii) $(x)_n = \sum_{k=1}^n \Delta(n,k) x^k$
 $(-1)^{n-k} c(n,k)$ (= signed) Stirling #'s of 1st kind

hence (iii) the infinite lower-triangular matrices

$$(S(n,k))_{\substack{n=1,2,\dots \\ k=1,2,\dots}} \text{ and } (\Delta(n,k))_{\substack{n=1,2,\dots \\ k=1,2,\dots}}$$

are inverses of one another,

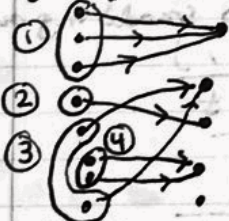
i.e., (iv) $\sum_{k=1}^n S(n,k) \Delta(k,m) = \delta_{n,m} = \sum_{k=1}^n \Delta(n,k) S(k,m)$
 Kronecker delta = $\begin{cases} 1 & \text{if } n=m \\ 0 & \text{otherwise} \end{cases}$

e.g. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 \\ 1 & 7 & 6 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & -3 & 1 & 0 \\ -6 & 11 & -6 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
 $\downarrow \quad \downarrow$
 $S(n,k) \quad \Delta(n,k)$

Pf: For (i), it is enough to prove when x is a nonnegative integer.
 (since it is an identity of polynomials...)

For $x=1, 2, 3, \dots$

$$x^n = \# \left\{ \begin{array}{c} \text{functions} \\ [n] \xrightarrow{f} [x] \end{array} \right\} = \sum_{\substack{\text{set partitions} \\ \pi \text{ of } [n]}} \# \left\{ \begin{array}{c} f: [n] \rightarrow [x] \text{ for which} \\ \text{set of preimages } \{f^{-1}(i)\}_{i \in [x]} \\ \text{(non-empty)} \end{array} \right\} = \pi$$



$\{1,2,3,4\}$
 $= \pi$ associated to f

$$= \sum_{k=1}^n S(n,k) \underbrace{x(x-1)(x-2)\dots(x-(k-1))}_{(x)_k}$$

choice of which $i \in [x]$ are
 images of the (non-empty)
 preimages determined by π

For (ii), recall that $x(x+1)(x+2)\dots(x+(n-1)) = \sum_{k=1}^n c(n,k) x^k$
 $x \mapsto (-x)$
 and mult. by $(-1)^n \rightsquigarrow x(x-1)(x-2)\dots(x-(n-1)) = \sum_{k=1}^n \Delta(n,k) x^k$

Then (iii) follows, because (i) and (ii) say that $S(n,k)$ and $\Delta(n,k)$
 are coeffs in transition between bases $\{x^n\}$ and $\{(x)_n\}$ of $\mathbb{C}[x]$.

10/8

The twelvefold way (Stanley §1.9)

By now we've seen many examples of counting how to put n balls into k boxes. The 12-fold way is a systematic approach to those kinds of problems, where:

- the balls can be distinguishable or indistinguishable,
- the boxes can also be dist. or indist.,
- the assignment of balls to boxes can be:
 - i) arbitrary,
 - ii) injective, i.e., at most one ball per box,
 - iii) surjective, i.e., at least one ball per box.

e.g. $n=3$
 $k=4$

$\begin{array}{|c|c|c|c|} \hline \textcircled{1}\textcircled{2} & & \textcircled{3} & \\ \hline 1 & 2 & 3 & 4 \\ \hline \end{array} \rightarrow \text{dist. balls/dist. boxes}$
 $\begin{array}{|c|c|c|c|} \hline \textcircled{1}\textcircled{2} & \textcircled{3} & & \\ \hline 1 & 2 & 3 & 4 \\ \hline \end{array} \rightarrow \text{dist. balls/indist. boxes}$
 $\begin{array}{|c|c|c|c|} \hline 00 & & 0 & \\ \hline 1 & 2 & 3 & 4 \\ \hline \end{array} \rightarrow \text{indist. balls/dist. boxes}$

Altogether, we get $12 = 2 \times 2 \times 3$ possibilities.

(Formally, we can view assignments as functions $f: [n] \rightarrow [k]$, and making balls/boxes indist. corresponds to 'modding out' by S_n/S_k action on domain/codomain.)

12-fold way	Any f	Injective f	Surjective f
Dist. Balls Dist. Boxes	1. k^n	2. $(k)_n = k(k-1)\dots(k-(n-1))$	3. $k! \cdot S(n, k)$
Indist. Balls Dist. Boxes	4. $\left(\begin{smallmatrix} k \\ n \end{smallmatrix} \right) = \binom{k+n-1}{n}$	5. $\binom{k}{n}$	6. $\left(\begin{smallmatrix} k \\ n-k \end{smallmatrix} \right) = \binom{n-1}{k-1}$
Dist. Balls Indist. Boxes	7. $\sum_{j=0}^k S(n, j)$	8. $\begin{cases} 1 & \text{if } n \leq k \\ 0 & \text{if } n > k \end{cases}$	9. $S(n, k)$
Indist. Balls Indist. Boxes	10. $\sum_{j=0}^k P_j(n) = P_k(n)$	11. $\begin{cases} 1 & \text{if } n \leq k \\ 0 & \text{if } n > k \end{cases}$	12. $P_k(n)$

Explanation of all these formulas:

1. (Dist balls + boxes, Any f): This very basic case was on the HW.
For each ball, we can choose one of k boxes $\rightarrow k^n$

2. (Dist balls + boxes, injective): Similar to previous case, but now choose boxes for balls $(1, 2, \dots, n)$ in order: first ball has k boxes, 2nd has $(k-1)$ because needs to be different from 1st, etc. $\rightarrow k(k-1)\dots(k-(n-1))$

3. (Dist balls + boxes, surjective): This determines an ordered set partition $(\pi_1, \pi_2, \dots, \pi_k)$ of $[n]$: $\begin{matrix} \textcircled{1,3} & \textcircled{5} & \textcircled{2,4} \\ \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} \end{matrix} \rightarrow \pi_1 = \{1, 3\}, \pi_2 = \{5\}, \pi_3 = \{2, 4\}$
of ordered s.p. of $[n]$ into k blocks $= k! \cdot \#$ (unordered) s.p. into k blocks
 $= k! \cdot S(n, k)$ why?

4. (Indist. balls / dist. boxes, any): This determines a weak composition of n into k parts: $\begin{matrix} \textcircled{0,0} & \textcircled{1} & \textcircled{0} & \textcircled{0,0,0} \\ \text{---} & \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} & \text{---} \end{matrix} \rightarrow * * | 1 * | * * *$
We've seen using stars + bars why $\binom{k+n-1}{n}$ is right answer.

5. (I. balls / d. boxes, inj.): Choose boxes that get a ball $\rightarrow \binom{n}{k}$

6. (I. balls / d. boxes, sur.): This determines a composition of n into k parts: $\begin{matrix} \textcircled{0,0} & \textcircled{0} & \textcircled{0,0} & \textcircled{0,0,0} \\ \text{---} & \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} & \text{---} \end{matrix} \rightarrow$ we saw before ans. $= \binom{n-1}{k-1}$.

7. (D. balls / I. boxes, any): This determines a set partition of $[n]$ into at most k blocks: $\begin{matrix} \textcircled{1,2} & \textcircled{3} & \textcircled{4,5} & \text{---} \\ \text{---} & \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} & \text{---} \end{matrix} \rightarrow S(n, 0) + S(n, 1) + \dots + S(n, k)$
(because some boxes may be empty!)

8. (D. balls, I. boxes, inj.): Only possibility here looks like: $\begin{matrix} \textcircled{1} & \textcircled{2} & \textcircled{3} & \dots & \textcircled{n} & \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} & \dots & \text{---} & \text{---} & \text{---} & \text{---} \end{matrix}$, which exists only if $n \leq k$.

10. (I. balls + boxes, any): This determines a partition of n into at most k parts: $\begin{matrix} \textcircled{0,0,0} & \textcircled{0,0,0} & \textcircled{0,0} & \textcircled{0,1} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \end{matrix}$
(because boxes may be empty!)

So if $P_k(n) = \#$ partitions of n into exactly k parts, answer is $P_0(n) + P_1(n) + P_2(n) + \dots + P_k(n) = P_k(n+k)$, as you saw on HW.

11. (I. balls + boxes, any): Only possibility: $\begin{matrix} \textcircled{0} & \textcircled{0} & \textcircled{0} & \dots & \textcircled{0} & \text{---} & \text{---} & \text{---} \\ \text{---} & \text{---} & \text{---} & \dots & \text{---} & \text{---} & \text{---} & \text{---} \end{matrix}$, which exists only if $n \leq k$.

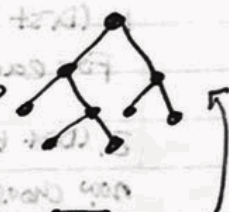
12. (I. balls + boxes, inj.): This is partition of n into k parts $\rightarrow P_k(n)$.

9. (D. balls / I. boxes, inj.): This is a set partition of $[n]$ into k blocks $\rightarrow S(n, k)$.

10/11

Catalan numbers (See Stanley's other book on this!)

$C_n := n^{\text{th}}$ Catalan number = # of plane binary trees w/ $n+1$ leaves (or n internal vertices, each w/ a L & R child)



= # of triangulations of $(n+2)$ -gon?



= # of lattice paths taking N E steps (weakly) $(0,0) \rightarrow (n,n)$, staying above diagonal $y=x$

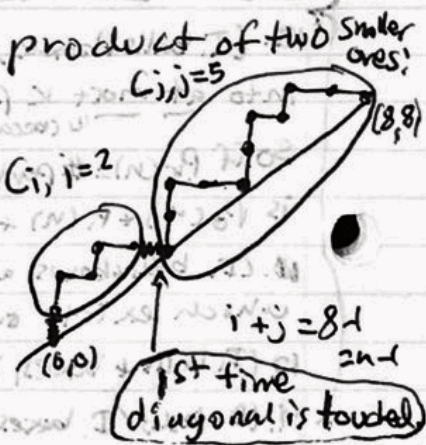
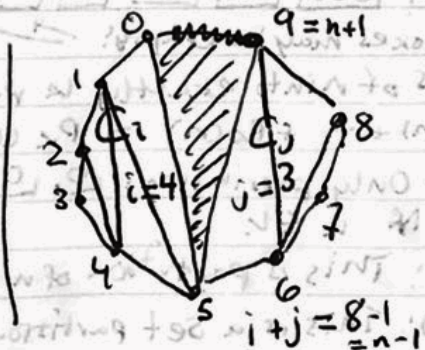
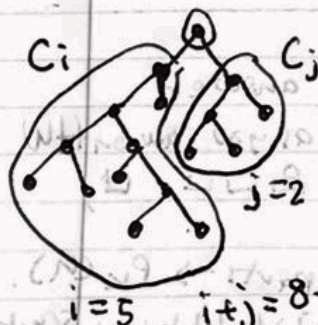


eg.	n	C_n	plane binary trees	triangulations	(lattice paths)
	0	1			
	1	1			
	2	2			
	3	5			

Prop. (fundamental recurrence)

$$C_n = \sum_{i+j=n-1} C_i \cdot C_j \quad \text{for } n \geq 1.$$

Pf: Each structure 'decomposes' as a product of two smaller ones!



Cor Setting $C(x) := \sum_{n \geq 0} C_n x^n$ we have
 $x C(x)^2 - C(x) + 1 = 0 \Rightarrow C(x) = \frac{1 \pm \sqrt{1-4x}}{2x}$ (quadratic thm)

Pf: fund. recurrence $C_n = \sum_{i+j=n-1} C_i \cdot C_j$ translates to

$$\begin{aligned} C(x) &= 1 + \sum_{n \geq 1} C_n x^n = 1 + \sum_{n \geq 1} \left(\sum_{i+j=n-1} C_i \cdot C_j \right) x^n \\ &= 1 + x \cdot \sum_{n \geq 0} \left(\sum_{i+j=n} C_i \cdot C_j \right) x^n \\ &= 1 + x \cdot C(x)^2. \end{aligned}$$

Thm $C_n = \frac{1}{n+1} \binom{2n}{n} \left(= \frac{(2n)!}{n!(n+1)!} = \frac{1}{2n+1} \binom{2n+1}{n} \right)$

Pf: Recall that $\frac{1}{\sqrt{1-4x}} = (1-4x)^{-1/2} = \sum_{n \geq 0} \binom{2n}{n} x^n$.

Integrate to get $-\frac{1}{2}(1-4x)^{1/2} = K + \sum_{n \geq 0} \frac{1}{n+1} \binom{2n}{n} x^{n+1}$

Check $x=0 \Rightarrow K = -\frac{1}{2} \Rightarrow$

$$\frac{1}{2} - \frac{1}{2}(1-4x)^{1/2} = \sum_{n \geq 0} \frac{1}{n+1} \binom{2n}{n} x^{n+1}$$

$$\frac{1 - \sqrt{1-4x}}{2x} = \sum_{n \geq 0} \frac{1}{n+1} \binom{2n}{n} x^n$$

Since $C(x) = \frac{1 - \sqrt{1-4x}}{2x}$, and $\frac{1 - \sqrt{1-4x}}{2x}$ has positive coeff's,

$$C(x) = \frac{1 - \sqrt{1-4x}}{2x} = \sum_{n \geq 0} \frac{1}{n+1} \binom{2n}{n} x^n \Rightarrow C_n = \frac{1}{n+1} \binom{2n}{n}. \quad \square$$

Can check $1 = \frac{1}{1} \binom{0}{0}$, $1 = \frac{1}{2} \binom{2}{1}$, $2 = \frac{1}{3} \binom{4}{2}$, $5 = \frac{1}{4} \binom{6}{3}$,
 and next term is $14 = \frac{1}{5} \binom{8}{4} = \frac{1}{5} \cdot 70$, so there
 are 14 triangulations of a hexagon (and 42 triang's of 7-gon)!